SEP490 · CAPSTONE PROJECT · REVIEW 2 · BM02

SRS/SDD Self-Review Report – Review 2

1. Run Information

Project code	FA26SE002 (group GFA26SE32)
Project name	Drone Delivery and Smart Locker Management System (DDSLMS) – product name Lock.R
Team	Lê Thị Yến Vi (SE182544, leader) · Trương Nguyễn Thái Bình (SE182370) · Nguyễn Quốc Bảo Huy (SE180211) · Trần Kim Nhã (SE182290)
Supervisor	Đặng Ngọc Minh Đức · Đỗ Hữu Tiến
AI tool and model	Claude Code (Anthropic) – model Claude Opus 5.5 (claude-opus-5-5)
Run date	09/11/2026
Repository tag / commit	Tag review-2 on github.com/LockR-Tech: backend fdf7480 · frontend 77cda59 · mobile b184a71 · iot f7bc145 · docs 8d2ac0b
Registration form version used	2.0 (proposed final: form signed 17/08/2026 with BM03 Record of Changes dated 09/11/2026)
Version	1.0
Date	09/11/2026

2. Standard Prompt

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You are reviewing the SRS and SDD of a software capstone project against its
registration form and source code. Report only; do not change any file.

Inputs: the registration form, SRS, SDD (with diagrams), source code and
database script in this repository.

Check:
1. Scope: FRs in the SRS not covered by the registration form, and
   registration items with no FR.
2. Requirements: FRs without an actor or testable acceptance criteria;
   NFRs without metric, threshold and measurement method.
3. Data: entities missing from the Logical ERD; ERD entities or attributes
   missing from the database script, and tables not in the ERD.
4. States: for each main entity, states used in code but missing from its
   state machine; transitions with no triggering FR; states with no exit.
5. Flows: main business flows in activity diagrams with no implementation
   or no test.
6. Architecture: components or third-party services in code but not in the
   architecture diagram, and the reverse.
7. Design vs code: modules, APIs or business rules implemented differently
   from the SDD.

Output one table: ID | Category | Location | Finding | Severity
(High/Medium/Low) | Evidence. Then list the 5 most important findings.
If something cannot be verified from the repository, say so; do not guess.
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3. Findings and Team Response

ID	Category	Location	Finding	Severity	Evidence	Team response
SR-01	Scope	SRS UC08, UC12, UC22; §3.2.24–3.2.26, §3.6.8	Wallet (top-up, payment by wallet, withdrawal to bank) is specified but is not in registration v2.0 and the mobile app no longer offers it.	High	mobile lib/ calls only GET /api/wallet/transactions, /spending-stats, /total-by-method; no top-up, wallet checkout or /api/wallet/withdraw call. payment-service WalletController still exposes them.	Fixed – customer wallet removed from SRS, SDD and Project Tracking, https://github.com/LockR-Tech/docs/commit/8771a1cb6212aff11997bc6914848827f805cd25
SR-02	Scope	SRS UC05	Face ID registration / sign-in is specified but is neither in the registration nor implemented.	Medium	mobile core/config/feature_flags.dart: faceRecognitionEnabled = false; backend has no /api/auth/ai/* endpoint.	Planned – remove Face ID from SRS – Huy, 15/11/2026
SR-03	Scope	Registration §3.2 c) Module 1 (ETA)	Registration item "Estimate delivery time (ETA)" has no FR in the SRS.	Medium	SRS contains no ETA requirement; order-service drone read model returns etaMinutes.	Planned – add ETA FR to SRS – Vi, 15/11/2026
SR-04	Scope	Registration §3.2 c) Module 3	Registration item "Generate maintenance notifications and alerts" has no FR in the SRS.	Medium	SRS §3.3.8, §3.4.5, §3.6.6 describe schedules and inspections only.	Planned – add maintenance alert FR to SRS – Nhã, 15/11/2026

ID	Category	Location	Finding	Severity	Evidence	Team response
SR-05	Scope	Registration §3.2 c) Modules 1–2	"Track delivery history and performance statistics" and "Monitor locker utilization" are only partly covered: UC25 lists revenue, orders by status, peak hours, user growth, top locations.	Low	SRS UC25, §3.6.2.	Planned – specify delivery statistics and utilization in UC24 – Bình, 15/11/2026
SR-06	Scope	SRS UC16; registration §3.2 c) Module 2	Opening an order's own box over Bluetooth is implemented, but it is neither in the registration nor specified in the SRS, which only has the admin-granted Bluetooth test (UC16).	Medium	iot-service BleOrderController POST /api/iot/ble/orders/{orderId}/token, BleOrderService, table ble_order_tokens; order-service BleOrderAccessService.	Planned – add Bluetooth order-open use case (G2) – Huy, 15/11/2026
SR-07	Scope	SRS UC08, UC09, UC13–UC16, UC24, UC28, UC31, UC34; §3.2.6, §3.2.16, §3.2.17, §3.4.6, §3.4.7	FRs not covered by registration v2.0: VietQR payment, hourly box rental, box incident tickets with SLA, vouchers and loyalty, Bluetooth open test, kiosk login by QR, parcel-drop recovery, favorite stores, delegate pickup / change PIN, ratings and complaints, mission planner, flight data.	Medium	SRS use cases and headings; registration v2.0 §3.2 c) Modules 1–4 do not mention them.	Kept – the registration keeps the signed scope; features added in Iteration 1 are specified in the SRS and listed in Design Brief §6
SR-08	Requirements	SRS §3.2–§3.6	FRs have no testable acceptance criteria: each function is described by screen, trigger and business rules. Testable criteria exist only for 16 features in Design Brief §6.	High	SRS has no acceptance-criteria field or section (no occurrence of "acceptance").	Planned – add acceptance criteria per FR (from Design Brief §6, G6) – Nhã, 22/11/2026
SR-09	Requirements	SRS §4.2.1–§4.2.4	NFRs give thresholds without a measurement method: availability 99 %, read < 1 s / write < 2 s, box open 5 s, assistant 10 s, usability 3 min / 30 s, capacity 100 sessions.	Medium	SRS §4.2 text.	Planned – add measurement method to each NFR (Design Brief §7) – Huy, 15/11/2026
SR-10	Data	Logical ERD (rp3/srs-assets/drawio/05*-logical-erd*; _generator/entities.json)	Seven tables in code and DB script are missing from the ERD: storage_handovers, drone_flight_blocks, drone_flight_track_points, drone_station_suspensions, drone_station_suspension_logs, ble_order_tokens, cabinet_hardware_configs.	Medium	@Entity StorageHandover, DroneFlightBlock, DroneFlightTrackPoint, DroneStationSuspension, DroneStationSuspensionLog, BleOrderToken, CabinetHardwareConfig; CREATE TABLE in Flyway migrations; absent from entities.json.	Planned – regenerate Logical ERD from current entities (G1) – Bình, 15/11/2026

ID	Category	Location	Finding	Severity	Evidence	Team response
SR-11	Data	Logical ERD	Ten columns added by migrations after the ERD was generated (29/10) are missing: orders.pending_extension_fee, orders.pending_extension_hours, lockers.deleted_at, locker_boxes.deleted_at, drone_units.deleted_at, user_profiles.deleted_at, user_profiles.language, user_profiles.terms_accepted, fcm_tokens.language, drone_parcel_incidents.recovery_station_dispatched_at.	Medium	ALTER TABLE … ADD COLUMN in Flyway migrations committed since 29/10/2026; absent from entities.json.	Planned – regenerate Logical ERD with new columns (G1) – Bình, 15/11/2026
SR-12	Data	Logical ERD; SDD §2	system_settings and system_setting_audits exist in the database of every service but are not in the ERD.	Low	Flyway migrations of auth, order, locker, payment, iot, store, loyalty and assistant services; common-lib business settings framework.	Planned – add settings tables to ERD – Bình, 15/11/2026
SR-13	States	Report 3, Report 4	Sixteen state machine diagrams exist only as draw.io sources; none is included in Report 3 or Report 4.	High	docs/rp3/srs-assets/drawio/06*-state-machine-*.drawio; SRS and SDD text has no state machine section.	Planned – insert state machines into Report 4 (G3) – Vi, 15/11/2026
SR-14	States	06f-state-machine-drone-delivery-session.drawio	State PARCEL_RELEASE_FAILED used in code is missing from the drone delivery session state machine.	Medium	DroneDeliverySession.PARCEL_RELEASE_FAILED; DroneLockerCommunicationService.processParcelReleaseFailed and handleCloseResult.	Planned – add PARCEL_RELEASE_FAILED to session state machine – Nhã, 15/11/2026
SR-15	States	SRS Table 5.4	Delivery stage values omit ACCEPTED, PICKUP_ASSIGNED, PICKED_UP, DROP_REPORTED, FAILED and CANCELED, which code sets (they are in diagram 06c).	Medium	order-service setDeliveryStage / advanceToStage values.	Planned – complete delivery stage values in SRS 5.4 – Vi, 15/11/2026
SR-16	States	SRS Table 5.4; 06-1-state-machine-order.drawio	REFUND_PENDING and REFUNDED are listed as order statuses in the SRS but are not in the order state machine; DELIVERY_INCOMPLETE is checked in code but never set and not modelled.	Low	DroneLockerCommunicationService CLOSED_ORDER_STATUSES (only occurrence of DELIVERY_INCOMPLETE).	Planned – align order statuses in SRS and diagram – Bình, 15/11/2026

ID	Category	Location	Finding	Severity	Evidence	Team response
SR-17	Flows	Report 4	Activity diagrams (06a send parcel, 06b rent locker, 06c drone delivery, 06d AI assistant, 06e knowledge base) exist only as draw.io sources and are not in Report 4.	High	docs/rp3/srs-assets/drawio/06a–06e-activity-*.drawio; SDD text has no activity diagram.	Planned – insert activity diagrams into Report 4 (G3) – Vi, 15/11/2026
SR-18	Flows	Design Brief A1 WF-04	Main flow "box incident to repair" has no activity diagram.	Medium	No activity diagram for incidents in docs/rp3/srs-assets/drawio or docs/diagrams.	Fixed – incident activity diagram 06f-activity-handle-incident-ticket.drawio added, https://github.com/LockR-Tech/docs/commit/e28e6c339a349f93b2a7a042a379615aa4a7dddc
SR-19	Flows	06c-activity-drone-delivery.drawio	The diagram shows the sender placing the parcel in the sending locker; code implements home pickup by a drone technician (offer, accept, pickup code and photo).	Medium	order-service DronePickupService; endpoints /api/drone-technician/drone-orders/{id}/pickup/*.	Planned – redraw drone activity diagram with home pickup (G3) – Vi, 15/11/2026
SR-20	Flows	Test suites; CI	No test runs a whole flow across services and clients; flows are tested per service. The admin web has no automated tests and the iot repository has no CI workflow. CI run results cannot be verified from the repository.	Medium	backend OrderServiceSendFlowTest, DroneHomePickupFlowPostgresTest; frontend has no test files and deploy.yml only builds; iot .github has no workflow.	Planned – map CI tests to §6 TC IDs, add cross-service e2e and iot CI (G5) – Nhã, 22/11/2026
SR-21	Architecture	SDD Figure 1.1, Table 1.1	The drone companion computer (drone-iot: ArUco camera, MJPEG stream, servo release, signed delivery events) is not an architecture component.	Medium	iot/drone-iot/lockr_drone/camera.py, delivery.py, release.py, mqtt_publisher.py.	Planned – add drone companion computer to architecture – Nhã, 15/11/2026
SR-22	Architecture	SDD Table 1.1 No. 22	eSMS.vn is an SMS provider in code but the architecture lists only Twilio. Whether SMS is enabled in production cannot be verified from the repository.	Low	notification-service EsmsSmsSender, SmsChannelConfig (provider chosen from environment).	Planned – add eSMS to architecture – Bình, 15/11/2026
SR-23	Architecture	SDD Table 1.1	Nominatim geocoding and Esri ArcGIS tiles used by the mobile app, and the landing page app, are not in the architecture.	Low	mobile pickup_geocoder.dart, location_picker.dart, drone_base_tile_layer.dart; frontend/landingPage.	Planned – add map services and landing page to architecture – Huy, 15/11/2026

ID	Category	Location	Finding	Severity	Evidence	Team response
SR-24	Design vs code	SDD §3.2; SRS UC15, §3.2.5	Kiosk login by QR: the kiosk UI never creates a QR session, so the designed flow cannot start.	Medium	iot/ui/src/api.js has no /api/auth/kiosk/qr-sessions call; auth-service KioskQrLoginController and mobile qr_login exist.	Planned – wire kiosk QR session or drop UC14 – Huy, 22/11/2026
SR-25	Design vs code	SDD Table 1.1 No. 03; SRS UC17; registration §3.1 Module 2	Pickup at the kiosk accepts a typed code only; the kiosk cannot scan a QR code.	Medium	iot/ui/src/api.js (unlockWithCode, unlockBox); no camera or QR library in the kiosk UI.	Planned – align SRS UC16 with registration change No. 6 (PIN at kiosk; Bluetooth or locker QR in the app) – Huy, 15/11/2026
SR-26	Design vs code	SDD §1.2.15; SDD drone sections	The SDD describes drone-iot as telemetry only; parcel release by servo (parcelRelease, DROP_PHASE_COMPLETED, PARCEL_RELEASE_FAILED) is implemented but not designed.	Medium	drone-iot lockr_drone/release.py, delivery.py; order-service DroneLockerCommunicationService; SDD has no mention of servo or parcel release.	Planned – design parcel release in SDD – Vi, 15/11/2026
SR-27	Scope	Registration §3.2 g) Phase 2; §3.3 b) In Scope	Registration item "Drone Simulator Service" has no FR; the SRS mentions only a payment simulator.	Medium	order-service DroneDeliverySimulator; SRS §3.2.10 (payment simulator only).	Planned – add demo simulator FR to SRS – Nhã, 15/11/2026
SR-28	Scope	Registration §3.2 b) Proposed Solutions	The registration states that drones operate between a central hub and Smart Locker stations; no FR covers a hub. The SRS specifies flights between two lockers with home pickup by a drone technician.	Medium	SRS UC10, §3.2.19, §3.4.1; order-service DroneSourceLockerSelector, DronePickupService.	Kept – the registration keeps the signed text; SRS and Design Brief describe the locker-to-locker model with home pickup

4. Top 5 Findings (as reported by the AI)

#	Finding ID	Summary
1	SR-08	SRS FRs have no testable acceptance criteria.
2	SR-13	State machine diagrams are not in Report 3/4.
3	SR-17	Activity diagrams are not in Report 4.
4	SR-01	Wallet is specified but out of registration and removed from the app.
5	SR-10	Seven tables in code are missing from the Logical ERD.

5. Summary

Severity	Total	Fixed	Kept	Planned
High	4	1	0	3
Medium	19	1	2	16
Low	5	0	0	5